3 Quantum statistical physics – single particles and waves

The problems are roughly in order of difficulty. Those indicated with ♡ are ones which are highly recommended. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

♡*Problem 3.1* **Internal energy and specific heat of an SHO**
Derive the internal energy, $U(T)$ and specific heat C for the **quantum** simple harmonic oscillator (note no distinction between different specific heats as volume is not a variable).

♡*Problem 3.2* **Spin waves in a ferromagnet**
The lowest lying excitations in magnets, where the crystal structure does not dictate special directions, are *spin waves*, where the spins precess around the average magnetisation direction with a phase difference between neighbouring sites.

Assume periodic boundary conditions in three dimensions with linear dimension L. The dispersion relation in a *ferromagnet* where all the spins are parallel in the ground state, is $\omega_{\mathsf{sw}}(\mathbf{k}) = Ak^2$, where A is sometimes called the *spin stiffness*. There is only *one* polarisation.[1]

Assume that the maximum wavevector is $|\mathbf{k}| \leq k_{\mathsf{m}}$. Derive an expression, as an integral, for the Helmholtz free energy, $F_{\mathsf{sw}}(T, V)$.

Problem 3.3 **Cosmic Microwave Background and a photon gas**
From the photon gas free energy, $F_{\mathsf{opt}}(T, V)$, derive the entropy, S, and the internal energy, U.

The progenitor of the Cosmic Microwave Background (CMB) was created about 3 minutes after the Big Bang when electromagnetic radiation decoupled from matter. (Roughly the Universe had cooled to a temperature such that $k_{\mathrm{B}}T < E_1$, the binding energy of the H atom.) To a good approximation there has been no exchange of heat with the remanent radiation since. Deduce a relationship between the temperature of the CMB and the "volume of the Universe".

Problem 3.4 **Contributions to the pressure within the Sun**
Both "particles" (electrons, nuclei etc) and photons contribute to the pressure in the Sun. Assume the particles are in the classical regime and form a single ideal gas.

Calculate the contributions from these two sources and their ratio in the following cases:

[1] This is perhaps surprising as the Hamiltonian which the spin waves arise from is the *Heisenberg model*, $\mathcal{H} = -\frac{1}{2}J\sum_{\mathbf{m},\mathbf{m}'} \mathbf{S}_{\mathbf{m}} \cdot \mathbf{S}_{\mathbf{m}'}$, where J is the exchange constant, and normally $\mathbf{m}$ and $\mathbf{m}'$ are neighbouring sites on a lattice. The spins are fixed-length vectors. $|\mathbf{S}| = \sqrt{S(S+1)}$, but there are still the *two* angular coordinates for the orientation of the spins. "It turns out" that roughly they are conjugate variables like x and p (actually $\cos\theta$ and ϕ), so there is only one degree of freedom and hence one mode.

(i) In the *Photosphere*, i.e. the Solar surface, where $T = 5800\,\mathrm{K}$ and the particle number density $N/V = 10^{23}$ particles m^{-3}.

(ii) In the centre of the Sun: $T = 1.5 \times 10^7\,\mathrm{K}$ and particle number density $N/V = 5 \times 10^{31}\,\mathrm{m}^{-3}$.

Problem 3.5 **Dust in the Solar System**
Consider dust with no angular momentum around the Sun. It experiences radiation pressure from the Sun's blackbody radiation and gravity due to the Sun. Ignore the planets.

Do large and small particles behave differently?

♡*Problem 3.6* **Adiabatic expansion of a photon gas**
Consider a photon gas in an adiabatic container of variable volume. Use the expression for the entropy of the gas, which is fixed, to deduce a relationship between p and V.

Problem 3.7 **Effective, or average, speed of sound.**
If the speeds of sound in a solid are two transverse modes with c_{tr} and one longitudinal mode with speed $\mathrm{c}_{\mathrm{long}}$. How should we define and average speed of sound, $\overline{\mathrm{c}}$, to enter the expression for the free energy of phonons in a solid in the same manner that the speed of light enters the free energy of light?